

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics

Practical – III TASK

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```
import pandas as pd
```

```
from sklearn.linear_model import LinearRegression
```

```
from sklearn.metrics import r2_score
```

```
data = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
```

Q.1 Find the correlation between Year_Built and House_Price.

```
correlation = data["Year_Built"].corr(data["House_Price"])
```

```
print("Correlation:", correlation)
```

```
Correlation: 0.051967361931064486
```

Q.2 Create a correlation matrix for all numerical columns.

```
print("\nCorrelation Matrix:")
```

```
print(data.corr(numeric_only=True))
```

```
Correlation Matrix:
               Square_Footage  Num_Bedrooms  Num_Bathrooms
Square_Footage           1.000000        -0.043564        -0.031584
Num_Bedrooms            -0.043564         1.000000         0.022848
Num_Bathrooms           -0.031584         0.022848         1.000000
Year_Built              -0.022392        -0.015820        -0.021600
Lot_Size                 0.089479        -0.009355         0.034923
Garage_Size              0.030593         0.113761         0.024846
Neighborhood_Quality    -0.008357        -0.049024         0.017500
House_Price              0.991261         0.014633        -0.001800

               Lot_Size  Garage_Size  Neighborhood_Quality
Square_Footage      0.089479      0.030593           -0.008357
Num_Bedrooms       -0.009355      0.113761           -0.049024
Num_Bathrooms       0.034923      0.024846           0.017500
Year_Built         -0.061050     -0.025485           -0.009357
Lot_Size            1.000000      0.002436           0.037630
Garage_Size         0.002436      1.000000           -0.011287
Neighborhood_Quali  0.037630     -0.011287           1.000000
House_Price         0.160412      0.052133           -0.007700
```

Q.3 Perform linear regression using:

X = Year_Built

Y = House_Price

Display the coefficient and intercept.

```
def regression_result(x_column):  
    X = data[[x_column]]  
    Y = data["House_Price"]  
    model = LinearRegression()  
    model.fit(X, Y)  
    prediction = model.predict(X)  
    r2 = r2_score(Y, prediction)  
    print("\nRegression:", x_column, "-> House_Price")  
    print("Coefficient:", model.coef_[0])  
    print("Intercept:", model.intercept_)  
    print("R2 Score:", r2)  
    return r2
```

```
Regression: Year_Built -> House_Price  
Coefficient: 638.6524884727304
```

Q.4 Calculate the R2 score for the regression model:

Year_Built - House_Price

```
r2_year = regression_result("Year_Built")
```

Q.5 Calculate the R2 score for:

Square_Footage - House_Price

```
r2_square = regression_result("Square_Footage")
```

Q.6 Calculate the R2 score for:

Neighborhood_Quality - House_Price

```
r2_quality = regression_result("Neighborhood_Quality")
```

Q.7 Compare the R2 scores of all three regression models.

```
print("\nR2 Score Comparison")  
print("Year_Built:", r2_year)  
print("Square_Footage:", r2_square)  
print("Neighborhood_Quality:", r2_quality)
```

```
R2 Score Comparison  
Year_Built: 0.0027006067060743044  
Square_Footage: 0.9825992634147099  
Neighborhood_Quality: 6.037338916142776e-05
```